Growing Pains

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Introduction

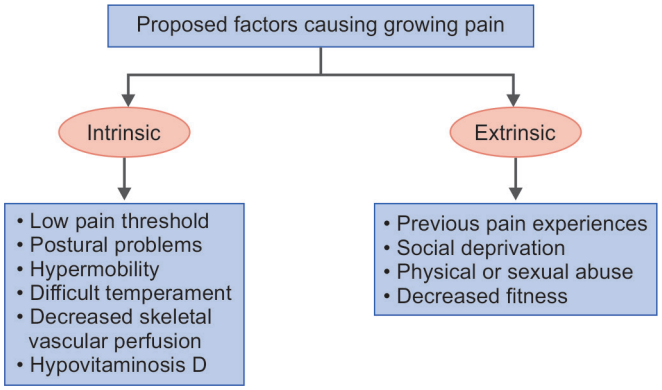
- ☑ Growing pains or benign musculoskeletal pain syndrome is one of the most common causes of recurrent pain in the preteens (<13 years of age).
- ☑ The estimated prevalence of growing pain ranges from 2% to 49.9% due to lack of diagnostic criteria adopted in different studies.
- ☑ Growing pains occurs in 10–20% of school age children.
- ☑ The peak age range is in between 3 and 8 years.
- ☑ No sex predilection was noted in the earlier studies.
- ☑ Joint hyper mobility and growing pain in school children, have a strong correlation.
- ☑ Children with growing pains complain of poorly localized pain in the calves and thighs, and not always localized to the joint.
- ☑ Pains tend to worsen in the evenings or nights and may follow a vigorous physical activity.
- ☑ 80–90% of patients have bilateral lower limb pains.
- ☑ Pains are often relieved with analgesics such as paracetamol or a simple massage.

Definition

The definition provided by Peterson simplifies the process of diagnosis.

- ☑ Intermittent pains once or twice per week
- ☑ Asymptomatic in between episodes
- ☑ Short lasting lower limb pains which are poorly localized
- ☑ Exclusion being nocturnal pain which continues next day morning or unilateral limb pain.

Definition of Growing Pains



Red Flag Signs

The presence of these should induce the Pediatrician to seek other diagnoses.

- ☑ Fever
- ☑ Weight loss
- ☑ Limp
- ☑ Inability to bear weight
- ☑ Regression of gross motor milestones
- ☑ Nocturnal pain which awakens the child
- ☑ Bony pain/tenderness
- ☑ Elevated ESR/CRP

Examination of a Child with Growing Pains

- ☑ Child is observed in the room for mood, play, chatter, mother's interaction with the child.
- ☑ Screening examination of all joints—*pGALS* screen (i.e., pediatric gait, arms, legs, spine). Presence of joint swelling or restriction could suggest a possibility of inflammatory arthritis. Hyperextensible joints would be suggestive of "benign joint hypermobility syndrome".
- ☑ Presence of allodynia or hyperesthesia would suggest a possibility of complex regional pain syndrome.
- ☑ Pallor, bruising, organomegaly or lymphadenopathy would be red flag signs.

Investigations

Lab investigations are generally unwarranted. But in case of presence of red flag signs, the following might be considered:

- ☑ Complete blood count and peripheral blood smear.
- ☑ Acute phase reactants (ESR, CRP)

Differential Diagnosis

- ☑ Benign joint hypermobility syndrome
- ☑ Juvenile idiopathic arthritis
- ☑ Complex regional pain syndrome (CRPS)
- ☑ *Infections:* Osteomyelitis, Septic arthritis, Cellultis and soft tissue abscess
- ☑ *Tumors:*
  - *Benign:* Osteoid osteoma and osteochondroma,
  - *Malignant:* Osteosarcoma, Ewing's sarcoma, leukemia and neuroblastoma
- ☑ Chronic non-bacterial osteomyelitis
- ☑ Trauma-accidental and non-accidental

Management

- ☑ Explanation and reassurance
- ☑ During bouts of pain, firm massaging and application of heat may comfort the child.
- ☑ Analgesia with paracetamol might be useful during episodes of pain.
- ☑ Children may benefit from:
  - *Exercise:* These include simple stretching exercises involving the quadriceps, hamstrings, gastrocnemius group of muscles.
  - Usually growing pains episodes show gradual decline in frequency over a period of few years and resolve by adolescence.

Summary

- ☑ Growing pain is a common and distressing clinical scenario in OPD practice.
- ☑ Careful clinical assessment and a sound knowledge together with application of the "rules" of growing pain, will help in making a diagnosis of growing pains in the majority of cases.
- ☑ Those justifying concern – usually the minority – will require investigation and possibly referral.
- ☑ Management in primary care requires reassurance, explanation and advice on relief of symptoms for the child.
- ☑ Finally, it is equally important to give clear instructions to the parents or care givers with indications for review, so that significant and potentially serious disease is not overlooked.

Further Reading

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Heart Failure

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